

EasyAcco

Platform Report

Full Feature History, Architecture & Product Roadmap

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Classification: Confidential — Internal Use Only
Prepared by: Engineering & Product Teams

HMRC 2026/27	UK SOLE TRADERS	FREE PLATFORM	CLIENT-SIDE
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SECTION 1

Platform Overview

EasyAcco is a UK freelancer tax calculation platform, built for the 2026/27 HMRC tax year. It is entirely free, requires no account to use core features, and runs all tax calculations client-side in the browser — no data ever leaves the device without explicit user consent.

Target Audience

UK sole traders, self-employed freelancers, directors, and anyone with self-assessment obligations. The platform is specifically scoped to the UK — multi-country support is out of scope by design.

Design System

The "System Auditor" aesthetic: quiet luxury, Nordic monochrome. All colour is derived from white on dark backgrounds via opacity modulation. Zero gold, navy, or lavender anywhere in the UI.

Token	Value	Usage
--sa-black	#181818	Page background
--sa-surface	#1C1D20	Card / panel background
--sa-gray	#222326	Input background
--sa-white	#F4F5F8	Primary text, buttons
--sa-muted	rgba(244,245,248,0.55)	Secondary text
--sa-border	rgba(244,245,248,0.07)	Dividers
--sa-green	#4ADE80	Success, positive values
--sa-red	#F87171	Error, negative values

SECTION 2

Past Updates — Commit History

All commits to the master branch, in chronological order from earliest to most recent.

Commit	Type	Description
abbb799	fix	Tax page responsive CSS leaking as visible text — removed stray CSS rules that were rendering as page content on mobile viewports
b91dabd	fix	Corrected 2026/27 student loan thresholds from official HMRC gov.uk data. Plan 2 corrected to £29,385; Plan 5 added at £25,000 threshold
91d54b4	fix	v4.0 — Corrected UI palette, removed dead CSS variables, fixed broken calculator logic, added 31 July Tax Pot feature
6107b90	feat	Brief v2 — New landing page, guest banner, GDPR compliance, updated 2026/27 tax allowances, editable Certificate of Self-Employment
a17ce8b	fix	Moved Kittax mascot component from global layout to AI page only — eliminates unnecessary asset loading on non-AI pages
398c79f	feat	Complete Kittax spec — Scotland toggle, student loans, HICBC, Married Couple's Allowance, professional AI response formatting
c2bd619	fix	Mobile sidebar, AI endpoint routing, UI polish + Supabase real-time sync implementation
3947aa5	fix	Desktop sidebar toggle behaviour + director NI pension salary sacrifice calculation correction
d372716	refactor	Site-wide palette migration: gold/navy/lavender removed; replaced with neutral dark System Auditor monochrome theme
3a02f3f	fix	Added .claude/ directory to .gitignore to prevent accidental token and session leakage to version control
af004fb	refactor	Phase 1 technical remediation (18 Apr 2026): unified tax engine, TypeScript migration, Tailwind refactor, 55 unit tests, Vercel deploy

SECTION 3

Current Architecture

As of 18 April 2026 — after Phase 1 technical remediation.

Library Layer (src/lib/)

File	Role	Status
tax-engine.ts	Unified barrel entry point — single import for all tax logic	NEW (Phase 1)
tax-logic.ts	Primary 2026/27 HMRC tax engine. Exports: calculateTax, validateTaxInput, round2, fmtGBP, calcPA, all types, RUK_TAXABLE_ADDITIONAL_THRESHOLD	Core
kittax-brain.ts	NLP conversational tax advisor with 25+ topic patterns. Now imports PA_BASE, PA_TAPER_START, RUK_TAXABLE_ADDITIONAL_THRESHOLD from tax-logic	Updated
TaxBible2026.ts	5 hard-coded HMRC scenario engines (employed, SE, high-earner, welfare, director). Now imports round2/fmtGBP from tax-logic	Updated
calculations.ts	Legacy simple estimator — superseded; DashboardUI now imports from tax-engine	Deprecated
__tests__/tax-engine.test.ts	55 Vitest unit tests covering all tax bands, NI, student loans, validation	NEW (Phase 1)

Application Layer (src/app/)

Route	File	Description
/	app/page.tsx	Landing page — Tailwind refactor complete (Phase 1), no inline C{} object
/dashboard	app/dashboard/DashboardUI.tsx	MTD calendar, quick estimator, nav tiles — updated to tax-engine
/dashboard/tax	app/dashboard/tax/page.tsx	Full TaxCalculator5 (35KB), all 5 scenarios, Scotland, student loan
/dashboard/expenses	app/dashboard/expenses/page.tsx	Category expense tracker with HMRC guidance
/dashboard/pnl	app/dashboard/pnl/page.tsx	P&L; income statement with monthly recharts
/dashboard/ai	app/dashboard/ai/page.tsx	Kittax AI advisor interface
/dashboard/currency	app/dashboard/currency/page.tsx	Live currency rates, 170+ currencies

/dashboard/transacti ons	app/dashboard/transactions/page.tsx	Double-entry ledger
/dashboard/learn	app/dashboard/learn/page.tsx	UK tax literacy learning cards
/api/ai/chat	app/api/ai/chat/route.ts	Claude API (Anthropic SDK) chat endpoint

Component Layer (src/components/)

Component	Language	Description
AppRouter.tsx	TypeScript	Navigation routes, Sidebar, NavLink — migrated from JSX (Phase 1)
CatMascot.tsx	TypeScript	Animated mascot — migrated from JSX, gold glow removed (Phase 1)
TaxCalculator5.tsx	TypeScript	35KB main tax calculator component
DemoDashboard.tsx	TypeScript	Guest/demo dashboard — migrated from JSX
TaxEstimator2026.tsx	TypeScript	Simplified estimator widget — migrated from JSX
CookieConsent.tsx	TypeScript	GDPR cookie consent banner
ui/Button.tsx	TypeScript	Base button component
ui/Card.tsx	TypeScript	Base card component
ui/Input.tsx	TypeScript	Base input component

SECTION 4

Tax Engine Constants 2026/27

All figures are HMRC 2026/27 compliant. Source file: src/lib/tax-logic.ts

Personal Allowance

Constant	Value	Notes
PA_BASE	£12,570	Standard personal allowance
PA_TAPER_START	£100,000	Taper begins here
PA_TAPER_END	£125,140	PA = £0 at this point
MARRIAGE_ALLOWANCE_XFER	£1,260	Transfer to partner (max saving £252)
BLIND_PERSONS_ALLOWANCE	£3,250	Additional PA for qualifying claimants

rUK Income Tax Bands

Band	Rate	Taxable Income Range
Basic Rate	20%	£0 — £37,700 (above PA)
Higher Rate	40%	£37,701 — £112,570 (above PA)
Additional Rate	45%	Above £112,570 (above PA) / gross above £125,140
60% Effective Rate	~60%	£100,000 — £125,140 gross (PA taper zone)

Constant Reconciliation (Phase 1 fix): kittax-brain.ts previously defined HIGHER_LIMIT = 112,570 (taxable-income basis). tax-logic.ts uses RUK_HIGHER_LIMIT = 125,140 (gross-income basis). Both are mathematically correct: 125,140 - 12,570 = 112,570. Phase 1 exported RUK_TAXABLE_ADDITIONAL_THRESHOLD from tax-logic and renamed the kittax-brain constant to TAXABLE_ADDITIONAL_THRESHOLD with a documented derivation.

Scotland Income Tax Bands

Band	Rate	Gross Income Ceiling
Starter Rate	19%	£16,537
Basic Rate	20%	£29,526
Intermediate Rate	21%	£43,662

Higher Rate	42%	£75,000
Advanced Rate	45%	£125,140
Top Rate	48%	Above £125,140

National Insurance

Class	Rate	Threshold	Applies To
Class 1 (main)	8%	£12,570 — £50,270	Employed / Directors (on salary)
Class 1 (upper)	2%	Above £50,270	Employed / Directors
Class 4 (main)	6%	£12,570 — £50,270	Self-Employed (on profit)
Class 4 (upper)	2%	Above £50,270	Self-Employed
Class 2 SPT	£0 (deemed)	Profit >= £7,105	Self-Employed — record protected
Class 2 Voluntary	£3.65/wk	Profit < £7,105	Optional to protect State Pension
Employer NI	15%	Salary above £5,000	Effective 6 April 2026 (was 13.8%)

Student Loan Repayment

Plan	Threshold	Rate	Eligible Group
Plan 1	£26,900	9%	Pre-2012 England/Wales
Plan 2	£29,385	9%	2012-2023 England/Wales
Plan 4	£33,795	9%	Scotland
Plan 5	£25,000	9%	Post-August 2023
Postgraduate	£21,000	6%	Postgrad Loan holders

Note: Student loan repayments are calculated on gross profit (before pension deductions). The engine uses grossProfit as the base, not taxableIncome — HMRC-correct behaviour.

Dividend Tax

Rate	Value	Band
Allowance	£500	Tax-free (2026/27)

Basic Rate	10.75%	Within basic rate band
Higher Rate	35.75%	Within higher rate band
Additional Rate	39.35%	Above higher rate band

Other Key Constants

Constant	Value
Pension Annual Allowance	£60,000 (capped at 100% of earnings)
MTD ITSA Threshold (from Apr 2026)	£50,000 gross self-employment income
MTD ITSA Extension (from Apr 2027)	£30,000 gross
VAT Registration Threshold	£90,000 taxable turnover
Redundancy Tax-Free Exemption	£30,000
Dividend Allowance	£500
ISA Annual Allowance	£20,000
CGT Annual Exempt Amount	£3,000
HICBC Threshold (starts)	£60,000
HICBC Full Clawback	£80,000

SECTION 5

Phase 1 Technical Remediation

Completed 18 April 2026. All items from the Chief Planner memorandum.

✓ Single Source of Truth

Created `src/lib/tax-engine.ts` as the unified barrel entry point. Removed duplicate `round2`, `fmtGBP`, and `calcPA` definitions from `TaxBible2026.ts` and `kittax-brain.ts`. All modules now import shared utilities from `tax-logic.ts`. `DashboardUI.tsx` migrated from stale `calculations.ts` (2024/25) to `tax-engine.ts` (2026/27).

✓ Reconcile Engine Constants

`HIGHER_LIMIT = 112,570` (taxable-income basis in `kittax-brain`) and `RUK_HIGHER_LIMIT = 125,140` (gross-income basis in `tax-logic`) were identified as the same boundary expressed in different bases. Resolution: `tax-logic.ts` now exports `RUK_TAXABLE_ADDITIONAL_THRESHOLD = RUK_HIGHER_LIMIT - PA_BASE = 112,570`. `kittax-brain` imports and uses this constant, eliminating the discrepancy.

✓ Enforce UI Standards & Refactor Styling

`src/app/page.tsx` refactored from inline `C{}` colour object to Tailwind utility classes using CSS custom properties (`var(--sa-*)`). `tailwind.config.js` updated: removed `gold`, `midnight`, `plum`, `lavender` colour tokens; replaced with `sa-*` palette matching `globals.css`. `CatMascot` gold glow (`rgba(253,185,49,...)`) replaced with white-tinted neutral shadow.

✓ Complete TypeScript Migration

`AppRouter.jsx` converted to `AppRouter.tsx` with proper TypeScript interfaces (`NavLinkProps`, typed `NAV_ITEMS`). `CatMascot.jsx` converted to `CatMascot.tsx` with `CatMascotProps` interface. `DemoDashboard.jsx` and `TaxEstimator2026.jsx` also converted. Zero `.jsx` files remain in `src/`. All components are strict TypeScript.

✓ Implement Core Engine Tests

Installed `Vitest` and `@vitest/ui`. Created `vitest.config.ts` with path aliases. Created `src/lib/__tests__/tax-engine.test.ts` with 55 unit tests. All 55 tests pass. TypeScript compiles with zero errors. Production build clean.

SECTION 6

Dashboard Modules

8 dashboard pages accessible after login or in guest mode.

1. Overview — /dashboard

MTD quarterly filing calendar (Q1–Q4 + Final), quick tax estimator widget, 8 navigation tiles. Urgency indicators: past (muted), upcoming (amber), urgent (red).

2. Tax Engine — /dashboard/tax

Full TaxCalculator5 component (35KB). All 5 HMRC employment scenarios. Scotland tax toggle. Student loan selector (Plans 1/2/4/5/Postgraduate). HICBC, Marriage Allowance, Blind Person's Allowance. Pension SIPP relief. Dividend tax stacking. Optimisation tips (60% trap escape, higher-rate pension, Ltd structure, ISA). Monthly breakdown. Step-by-step audit trail.

3. Expenses — /dashboard/expenses

Categorised expense tracking with HMRC "wholly and exclusively" guidance. Business category classification.

4. Ledger — /dashboard/transactions

Double-entry transaction log. Net position summary at a glance.

5. P&L; Reports — /dashboard/pnl

Income statement. Monthly Recharts visualisation. MTD-ready export.

6. Currency — /dashboard/currency

Live exchange rates for 170+ currencies. Useful for international freelancer invoicing.

7. Learn — /dashboard/learn

UK tax literacy cards from src/content/learning-cards.json. Covers tax year history, allowances, and HMRC rules.

8. Tax Advisory — /dashboard/ai

Kittax AI NLP advisor. 25+ topic patterns: income calc, PA taper, 60% trap, dividends, pensions, expenses, mileage, NI, student loans, ISA, CGT, VAT, MTD, deadlines, redundancy, welfare, HICBC, Marriage Allowance, trading losses.

Phase 2 Roadmap — Tier 1

High-value retention features. Transforms the app from a one-off utility into a weekly-use dashboard. **Immediate Backlog — next sprint after Phase 1 merges.**

Feature	Description	Strategic Goal
"Set Aside for Tax" Calculator	Real-time calculation of exactly how much a user needs to put into a separate savings pot from their current month's income. Factors in income type, expenses, and student loan.	Solves the #1 freelancer pain point. Reduces January tax shock.
Year-Round Tax Tracker	Dashboard aggregating logged income/expenses via Supabase to project the January 2028 Self Assessment tax bill in real-time. Running total updates as transactions are logged.	Drives weekly active usage — users return to log income and check their running liability.
Scenario Comparison	Side-by-side UI comparing financial outcomes across configurations (e.g., Sole Trader vs Ltd Company) with colour-coded deltas for each line item.	High-value decision support. Helps users decide whether incorporating is worth it for their income level.
Invoice Lifecycle Tracking	Expand the current invoice generator to track sent, paid, and overdue statuses. Include one-click chase-email templates and a payment calendar.	Creates recurring platform dependency — users must return to update invoice status.

SECTION 8

Phase 2 Roadmap — Tier 2

Domain-depth features. Designed to replace expensive manual accounting workflows. **Mid-Term Backlog.**

Feature	Description	Strategic Goal
Self Assessment Pre-filler	Mapping engine outputs directly to SA100 form box numbers (e.g., "Enter £X in Box 3.29"). Generates a complete cheat sheet to fill in the HMRC return. No actual filing integration.	Direct cost-saving — replaces the need for an accountant for straightforward returns.
Receipt OCR Upload	Use Vision APIs (Tesseract.js already installed) to automatically parse amount, vendor, and date from snapped receipt photos. Auto-populates expense records.	Eliminates the #1 friction in expense tracking — manual data entry.
Category Insights	Flagging anomalies and optimisation opportunities. E.g., "Your home office claim is £200 but the simplified flat rate would give you £312 — consider switching."	Demonstrates immediate, specific ROI to the user.
Automated Deadline Calendar	Visual countdown timers for all key HMRC dates (31 Jan, 31 Jul, quarterly MTD). Email reminders 7 days prior. Opt-in email capture.	Captures user email addresses (growth asset) while providing genuine utility.

SECTION 9

Phase 2 Roadmap — Tier 3

Polish & Growth. Low-effort, high-impact features for user acquisition and UX comfort.

Transparency & Verification Page

A dedicated page with side-by-side screenshots proving the EasyAcco engine's accuracy against HMRC's official tax calculator. Provides trust signals for new users who are sceptical of third-party tools.

1-Click Referral

Pre-filled WhatsApp message and email share button for organic word-of-mouth growth among freelancer communities. Minimal implementation — one button, pre-composed message.

Dedicated Learning Hub

Expand `src/content/learning-cards.json` into permanent, SEO-optimised long-form articles. Targets evergreen search traffic (e.g., "How to register for VAT UK", "What is the 60% tax trap"). No additional content management system needed — markdown files compiled at build time.

Light/Dark Mode Toggle

User preference toggle for light or dark mode while maintaining the strict monochrome design system. Light mode uses inverted palette (near-white background, near-black text). Preference persisted to `localStorage`.

SECTION 10

Do Not Build List

Strictly out of scope. These decisions are final for the foreseeable future to maintain focus.

■ NO Mobile App

A responsive web application is sufficient. Native apps require App Store logistics, separate codebases, review cycles, and platform-specific APIs. Engineering effort is not justified given the web app's full mobile responsiveness.

■ NO Multi-Country Support

EasyAcco wins by being the undisputed best tool for UK freelancers. Adding US, Canadian, or Australian tax logic dilutes the product, confuses the user base, and requires maintaining multiple sets of constantly-changing tax regulations.

■ NO Subscriptions or Paywalls (Yet)

Growth and retention are the only metrics that matter in the current phase. The goal is to build a loyal, active user base on a free product before introducing any monetisation. Introducing a paywall now would kill user acquisition.

■ NO Forums or Communities

Moderation is a massive, ongoing resource drain. The freelancer community already has active subreddits (r/freanceuk, r/UKPersonalFinance) and Facebook groups. Point users there rather than competing.

■ NO Agentic AI

The AI (Kittax) remains a helpful, read-only advisor. Do not attempt to build a system that automatically files returns, communicates with HMRC, or takes financial actions on the user's behalf. Keep AI simple and deterministic — it reduces liability and user anxiety.

Deployment & Infrastructure

Property	Value
Platform	Vercel (auto-deploy via GitHub integration)
Latest Deployment	easyacco-67iwtjvph-fionadbarads-projects.vercel.app
Deployment ID	dpl_3uPPbqGfQH9LckiUkuutYAt12p2u
Deployment Date	18 April 2026
Repository	github.com/fionadbarad/easyaccos
Branch	master
Build Command	next build --webpack
Framework	Next.js 16.2.3
React Version	19.2.4
TypeScript	6.0.2
Tailwind CSS	4.2.2
Database	Supabase (PostgreSQL with RLS)
Auth	Supabase Auth + JWT
AI	Anthropic Claude API (@anthropic-ai/sdk ^0.82.0)
Testing	Vitest 4.1.4
Analytics	@vercel/analytics, @vercel/speed-insights

Environment Variables Required

Variable	Purpose
NEXT_PUBLIC_SUPABASE_URL	Supabase project URL
NEXT_PUBLIC_SUPABASE_ANON_KEY	Supabase anonymous (public) key
ANTHROPIC_API_KEY	Claude API key for AI advisor endpoint

SECTION 12

Test Coverage Summary

All tests are in `src/lib/__tests__/tax-engine.test.ts`. Run with: **npm test**

55 / 55 tests passing — Zero failures

Test Suite	Count	Cases Covered
Reconciled constants	1	<code>RUK_TAXABLE_ADDITIONAL_THRESHOLD = RUK_HIGHER_LIMIT - PA_BASE = 112,570</code>
round2 utility	4	Rounding, zero, truncation, float drift elimination ($0.1+0.2=0.3$)
Personal Allowance taper	7	Below threshold, at £110k, at £125,140, above £125,140, £100,001, £100,002
rUK income tax — basic	3	No tax at PA, basic rate only, basic band exhaustion at £50,270
rUK income tax — higher/addl	3	Higher rate trigger, additional rate at £150k, 60% trap flag
60% trap flag	3	Active at £110k, inactive below £100k, inactive at £125,140+
Scotland income tax	2	Starter rate band, no tax below PA
NI Class 4 (self-employed)	3	Below PT, main rate, upper rate (split at UEL)
NI Class 1 (employed)	2	8% rate, no Class 4 for employed
NI Class 2	3	Deemed above SPT (£7,105), not deemed below, voluntary opt-in
Student Loan — all plans	6	Plans 1/2/4/5/Postgrad + base=grossProfit (before pension)
Pension contributions	3	Reduces income, £60k cap, capped at grossProfit
Dividend tax	2	£500 allowance (tax-free), 10.75% basic rate on excess
Special allowances	2	Marriage (-£1,260 PA), Blind Persons (+£3,250 PA)
Allowable expenses	2	Reduces grossProfit, negative profit clamped to zero
Input validation	5	Zero revenue, $\text{expenses} \geq \text{revenue}$, negative dividends, $\text{pension} > £60k$, valid
MTD warning	2	Fires above £50,000 gross, does not fire at exactly £50,000
Monthly breakdown	1	$\text{monthly} \times 12 \approx \text{annual}$ within rounding tolerance
Effective tax rate cap	1	Never exceeds 100% (tested at £500,000)

The test suite validates every tax band boundary, NI threshold, student loan plan, edge case, and validation rule. It serves as the living specification for the tax engine — any regression in the 2026/27 HMRC calculations will be caught immediately.

End of Report

EasyAcco — UK Freelancer Tax Platform — Not Financial Advice